

libraries, the main facet is that their height is fixed- meaning, they could be placed in rows, thereby making the process of automated digital layout easier. Generally, the cells are optimized full-custom layouts. This means delays as also the surface area would be minimized.

Two of the key components that can be found in a typical standard-cell library are given below:

1. Library Database

This database contains multiple views. These often include schematic, symbol, layout, abstract and also other logical or simulation views. Using the database, different information could get captured in different formats including the Synopsys Milkyway and cadence LEF formats. These would have reduced information regarding the layouts which would be enough for automated “Place and Route” tools.

2. Timing Abstract

This abstract gives functional definitions as well as information regarding power, timing and noise for every cell. Usually, the abstract is found in Liberty format.

Some additional components that could be found in a standard-cell library are given here:

- The complete layout of the cells
- The cells’ spice models
- Verilog models
- Parasitic Extraction models
- DRC rule decks

Application

Now, let’s take a look at the application of standard cell.

It’s safe to say that a 2-input NOR or NAND function is more than enough to form any arbitrary Boolean function set. However, in modern ASIC design, there’s a general practice of using standard-cell methodology using a sizable library- or maybe multiple libraries-of cells.

Usually, the library consists of multiple implementation of the same logic function. The differences would be in speed and the area it takes up. The variety has the advantage that it improves the efficiency of the tools for automated synthesis, place and route (SPR). Also, the designer gets a higher freedom to perform implementation trade-offs.